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Display substrate and display device

Abstract

A display substrate and a display device are provided. The display substrate includes a base substrate, at least one group of contact pads, a plurality of first light-emitting elements, a plurality of first pixel driving circuits, a plurality of connecting traces, a plurality of data lines and a plurality of leads. The display area includes a first display area and a second display area; the first light-emitting elements are located in the first display area; the first pixel driving circuits are located in the second display area; the leads are located in the first display area and the peripheral area, and connect the data lines and the at least one group of contact pads; orthographic projections of the first light-emitting elements on a substrate surface of the base substrate are at least partly overlapped with orthographic projections of the leads on the substrate surface of the base substrate.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) This application is the National Stage of PCT/CN2021/091654 filed on Apr. 30, 2021, the disclosure of which is incorporated by reference.
TECHNICAL FIELD
(2) Embodiments of the present disclosure relate to a display substrate and a display device.
BACKGROUND
(3) For OLED (Organic Light-Emitting Diode) display products, there are many types of circuit units, and each of these circuits plays its own role in displaying. At present, the display products are developing in the direction that a bezel around a display area is continuously narrowed. Especially, a lower bezel of the display area occupies a larger space due to a fan-out area (for example, a lead for connecting a circuit of the display area).
SUMMARY
(4) At least one embodiment of the present disclosure provides a display substrate. The display substrate includes: a base substrate, including a display area and a peripheral area located at least at one side of the display area, the display area including a first display area and a second display area, and the first display area being located between the second display area and the peripheral area in a first direction; at least one group of contact pads located in the peripheral area; a plurality of first light-emitting elements located in the first display area; a plurality of first pixel driving circuits located in the second display area; a plurality of connecting traces, one end of at least one connecting trace of the plurality of connecting traces being electrically connected with at least one first light-emitting element of the plurality of first light-emitting elements, and the other end of the at least one connecting trace being electrically connected with the first pixel driving circuit; a plurality of data lines located in the second display area and configured to provide data signals to the plurality of first pixel driving circuits; and a plurality of leads located in the first display area and in the peripheral area, the plurality of leads connecting the plurality of data lines and the at least one group of contact pads; wherein orthographic projections of the plurality of first light-emitting elements on a substrate surface of the base substrate are at least partly overlapped with orthographic projections of the plurality of leads on the substrate surface of the base substrate.
(5) For example, in the display substrate provided by at least one embodiment of the present disclosure, the orthographic projections of at least some first light-emitting elements of the plurality of first light-emitting elements on the substrate surface of the base substrate have no overlap with orthographic projections of the plurality of first pixel driving circuits and the plurality of data lines on the substrate surface of the base substrate.
(6) For example, the display substrate provided by at least one embodiment of the present disclosure further includes a plurality of second pixel driving circuits and a plurality of second

light-emitting elements; wherein the plurality of second pixel driving circuits and the plurality of second light-emitting elements are located in the second display area, and the plurality of second pixel driving circuits are configured to drive the plurality of second light-emitting elements to emit light; the plurality of second pixel driving circuits are electrically connected with the plurality of data lines, and the plurality of data lines are configured to provide data signals to the plurality of second pixel driving circuits; and orthographic projections of the plurality of second light-emitting elements on the substrate surface of the base substrate are at least partly overlapped with orthographic projections of the plurality of second pixel driving circuits on the substrate surface of the base substrate.
(7) For example, in the display substrate provided by at least one embodiment of the present disclosure, the plurality of second pixel driving circuits include a plurality of second pixel circuit groups arranged along the first direction and extending along a second direction, the first direction intersects with the second direction, and each of the plurality of second pixel circuit groups includes a plurality of rows of second pixel driving circuits extending along the second direction; the plurality of first pixel driving circuits include a plurality of first pixel circuit groups arranged along the first direction and extending along the second direction, and each of the plurality of first pixel circuit groups includes a plurality of rows of first pixel driving circuits extending along the second direction; and the plurality of rows of second pixel driving circuits are located at a side of the plurality of rows of first pixel driving circuits away from the first display area.
(8) For example, the display substrate provided by at least one embodiment of the present disclosure further includes at least one row of first dummy pixel driving circuits extending along the second direction; in the first direction, the at least one row of first dummy pixel driving circuits are distributed at intervals between the at least one row of second pixel driving circuits extending along the second direction.
(9) For example, in the display substrate provided by at least one embodiment of the present disclosure, in the first direction, the first dummy pixel driving circuit is not provided between a plurality of rows of first pixel driving circuits extending in the second direction.
(10) For example, in the display substrate provided by at least one embodiment of the present disclosure, in the first direction, two adjacent rows of first dummy pixel driving circuits extending in the second direction are spaced apart by at least one second pixel circuit group.
(11) For example, in the display substrate provided by at least one embodiment of the present disclosure, a size of the second pixel driving circuit in the first direction is larger than that of the first pixel driving circuit in the first direction.
(12) For example, in the display substrate provided by at least one embodiment of the present disclosure, the size of the second pixel driving circuit in the first direction is in the range from 63 microns to 65 microns, and the size of the first pixel driving circuit in the first direction is in the range from 60 microns to 62 microns.
(13) For example, in the display substrate provided by at least one embodiment of the present disclosure, a size of the first pixel driving circuit and a size of the second pixel driving circuit are equal in the first direction.
(14) For example, in the display substrate provided by at least one embodiment of the present disclosure, the size of the first pixel driving circuit and the size of the second pixel driving circuit are equal in the first direction, and have a value in the range from 63.8 microns to 63.9 microns.
(15) For example, in the display substrate provided by at least one embodiment of the present disclosure, the plurality of connecting traces include a first connecting trace partially located in the first display area and a second connecting trace located in the second display area; wherein the second connecting trace is electrically connected with a corresponding second light-emitting element, the first connecting trace is electrically connected with a corresponding first light-emitting element, and the corresponding second light-emitting element electrically connected with the second connecting trace and the corresponding first light-emitting element electrically connected

with the first connecting trace emit light with the same color; and a line width of the second connecting trace is larger than that of the first connecting trace.
(16) For example, the display substrate provided by at least one embodiment of the present disclosure further includes a first metal layer, a second metal layer and a third metal layer, wherein the first metal layer is located on the base substrate, the second metal layer is located at a side of the first metal layer away from the base substrate, and the third metal layer is located at a side of the second metal layer away from the base substrate; the plurality of leads are arranged in the same layer as the first metal layer or the second metal layer; or some leads of the plurality of leads are arranged in the same layer as the first metal layer, and the other leads of the plurality of leads are arranged in the same layer as the second metal layer; and the some leads and the other leads are alternately arranged.
(17) For example, in the display substrate provided by at least one embodiment of the present disclosure, the peripheral area further includes a bending area which is located at a side of the first display area away from the second display area, and the bending area includes a plurality of signal lines, the plurality of signal lines extend along the first direction and are connected in one-to-one correspondence with the plurality of leads; wherein the first direction intersects with the second direction.
(18) For example, in the display substrate provided by at least one embodiment of the present disclosure, the plurality of leads include a plurality of first leads and a plurality of second leads, the plurality of first leads are connected in one-to-one correspondence with the plurality of second leads, the plurality of first leads are located in the first display area, and the plurality of second leads are located in the peripheral area and are connected in one-to-one correspondence with the plurality of signal lines.
(19) For example, in the display substrate provided by at least one embodiment of the present disclosure, the plurality of data lines have a first average line spacing in the second direction, the plurality of signal lines have a second average line spacing in the second direction, and the plurality of first leads have a third average line spacing in the second direction; wherein the third average line spacing is between the first average line spacing and the second average line spacing.
(20) For example, in the display substrate provided by at least one embodiment of the present disclosure, the third average line spacing of the plurality of first leads is decreased, in the first direction, from a side close to the plurality of data lines to another side away from the plurality of data lines.
(21) At least one embodiment of the present disclosure provides a display device, including the display substrate described in any of the above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In order to clearly illustrate the technical solutions of the embodiments of the disclosure, the drawings of the embodiments will be briefly described in the following; it is obvious that the described drawings are only related to some embodiments of the disclosure and thus are not limitative to the disclosure
(2) FIG. **1** is a schematic diagram of a display substrate;
(3) FIG. **2** is a schematic diagram of a display substrate provided by at least one embodiment of the present disclosure;
(4) FIG. **3**A is a schematic diagram of another display substrate provided by at least one embodiment of the present disclosure;
(5) FIG. **3**B is a schematic diagram of yet another display substrate provided by at least one embodiment of the present disclosure;

(6) FIG. **4** is a partial structural diagram of a first display area of a display substrate provided by at least one embodiment of the present disclosure;
(7) FIG. **5**A is a partial structural diagram of a pixel driving circuit of a display substrate provided by at least one embodiment of the present disclosure;
(8) FIG. **5**B is a partial structural diagram of a pixel driving circuit and a light-emitting element of a display substrate provided by at least one embodiment of the present disclosure;
(9) FIG. **5**C is a schematic diagram of a pixel circuit group of a display substrate before and after compressed as provided by at least one embodiment of the present disclosure;
(10) FIG. **6**A is another partial structural diagram of a pixel driving circuit of a display substrate provided by at least one embodiment of the present disclosure;
(11) FIG. **6**B is another partial structural diagram of a pixel driving circuit and a light-emitting element of a display substrate provided by at least one embodiment of the present disclosure;
(12) FIG. **7** is a schematic cross-sectional view of a display substrate provided by at least one embodiment of the present disclosure; and
(13) FIG. **8** is a schematic diagram of a display device provided by at least one embodiment of the present disclosure.

DETAILED DESCRIPTION

(14) In order to make objects, technical details and advantages of the embodiments of the disclosure apparent, the technical solutions of the embodiments will be described in a clearly and fully understandable way in connection with the drawings related to the embodiments of the disclosure. Apparently, the described embodiments are just a part but not all of the embodiments of the disclosure. Based on the described embodiments herein, those skilled in the art can obtain other embodiment(s), without any inventive work, which should be within the scope of the disclosure.
(15) Unless otherwise defined, all the technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art to which the present disclosure belongs. The terms "first," "second," etc., which are used in the present disclosure, are not intended to indicate any sequence, amount or importance, but distinguish various components. Also, the terms "comprise," "comprising," "include," "including," etc., are intended to specify that the elements or the objects stated before these terms encompass the elements or the objects and equivalents thereof listed after these terms, but do not preclude the other elements or objects. The phrases "connect", "connected", etc., are not intended to define a physical connection or mechanical connection, but may include an electrical connection, directly or indirectly. "On," "under," "right," "left" and the like are only used to indicate relative position relationship, and when the position of the object which is described is changed, the relative position relationship may be changed accordingly.
(16) FIG. **1** is a schematic diagram of a display substrate.
(17) For example, as shown in FIG. **1**, the display substrate **01** includes a display area AA and a peripheral area **012** surrounding the display area AA. The peripheral area **012** includes a lower bezel located at one side of the display area AA. The lower bezel includes a fan-out area FNT and a bonding area COF. The fan-out area FNT is located between the display area AA and the bonding area COF. The bonding area COF is configured to be bonded with a signal input element, for example, the signal input element includes an integrated circuit (IC), and for another example, the signal input element includes a data driving circuit IC. The fan-out area FNT includes a plurality of traces electrically connected with signal lines located in the display area AA, and the plurality of traces are also electrically connected with signal input elements of the bonding area COF. The display area AA includes a plurality of sub-pixels arranged in an array and configured to display an image. Because the fan-out area takes up a larger space, the lower bezel of the display substrate **01** is difficult to be narrowed.
(18) For example, as shown in FIG. **1**, the display substrate **01** includes a plurality of pixel driving circuits **010** arranged in an array. The display area AA may include a plurality of light-emitting

elements (not shown in the figure) arranged in an array. The plurality of light-emitting elements are electrically connected with the plurality of pixel driving circuits **010**, for example, in one-to-one correspondence, and a periodic size of the pixel driving circuits **010** is as same as that of the light-emitting elements, under general conditions. At present, it is difficult to narrow the lower bezel of the display substrate **01**.

(19) At least one embodiment of the present disclosure provides a display substrate, which includes a base substrate, at least one group of contact pads, a plurality of first light-emitting elements, a plurality of first pixel driving circuits, a plurality of connecting traces, a plurality of data lines and a plurality of leads. The base substrate includes a display area and a peripheral area located at least at one side of the display area; the display area includes a first display area and a second display area, and the first display area is located between the second display area and the peripheral area in a first direction; the at least one group of contact pads is located in the peripheral area; the plurality of first light-emitting elements are located in the first display area; the plurality of first pixel driving circuits are located in the second display area; one end of at least one connecting trace among the plurality of connecting traces is electrically connected with at least one first light-emitting element, and the other end of the at least one connecting trace is electrically connected with the first pixel driving circuit; the plurality of data lines are located in the second display area and configured to provide data signals to the plurality of first pixel driving circuits; the plurality of leads are located in the first display area and in the peripheral area, and connect the plurality of data lines and the at least one group of contact pads; orthographic projections of the plurality of first light-emitting elements on a substrate surface of the base substrate are at least partially overlapped with orthographic projections of the plurality of leads on the substrate surface of the base substrate.

(20) At least one embodiment of the present disclosure also provides a display device including the display substrate described above.

(21) In the display substrate and display device provided by the above embodiments, the orthographic projections of the plurality of first light-emitting elements on the substrate surface of the base substrate are at least partly overlapped with the orthographic projections of the plurality of leads on the substrate surface of the base substrate, that is, the plurality of leads are moved upwardly in a direction close to the second display area along the first direction, and at least some of the plurality of leads are overlapped with the first display area, so that at least some of the plurality of leads are arranged at a side of the first light-emitting elements close to the base substrate in a direction perpendicular to the substrate surface of the base substrate, thereby narrowing the lower bezel of the display substrate.

(22) Embodiments and examples of the present disclosure will be described in detail below with reference to the accompanying drawings.

(23) FIG. **2** is a schematic diagram of a display substrate provided by at least one embodiment of the present disclosure; FIG. **3**A is a schematic diagram of another display substrate provided by at least one embodiment of the present disclosure; and FIG. **4** is a partial structural diagram of a first display area of a display substrate provided by at least one embodiment of the present disclosure.

(24) For example, in some embodiments, as shown in FIGS. **2** and **3**A, the display substrate **1** includes a base substrate **100**. The base substrate **100** includes a display area **10** and a peripheral area **20**. The display area **10** is, for example, a gray translucent area as shown in FIG. **3**A. The display area **10** includes a first display area **101** and a second display area **102**. In the first direction Y (for example, the longitudinal direction in the figure), the first display area **101** is located between the second display area **102** and the peripheral area **20**. For example, the peripheral area **20** surrounds the first display area **101** and the second display area **102**. The peripheral area **20** may also be located at one side of the first display area **101** or the second display area **102**. For example, in the first direction Y, the first display area **101** is closer to the peripheral area **20**. For example, in some embodiments, as shown in FIG. **2**, the display substrate **1** further includes at least one group of contact pads BD**1** located in the peripheral area **20**. For example, the peripheral area **20** includes

a bonding area **21** in which the at least one group of contact pads BD**1** are arranged along the second direction X. The at least one group of contact pads BD**1** is configured to be bonded with a signal input element, for example, the signal input element includes an integrated circuit (IC), and for another example, the signal input element includes a data driving circuit IC. For example, in the first direction Y, the first display area **101** is closer to the at least one group of contact pads BD**1** than the second display area **102**.
(25) For example, as shown in FIGS. **3**A and **4**, the display substrate **1** further includes a plurality of first light-emitting elements **11**A (as shown in FIG. **4**). The plurality of first light-emitting elements **11**A are located in the first display area **101**. For example, the display area **10** shown in FIG. **3**A includes a plurality of light-emitting elements arranged in an array. Among the plurality of light-emitting elements, the light-emitting elements located in the first display area **101**, for example, are referred to as first light-emitting elements **11**A. For example, the first light-emitting elements **11**A may include a first light-emitting element **1101**A emitting blue light, a first light-emitting element **1102**A emitting green light, a first light-emitting element **1103**A emitting red light and a first light-emitting element **1104**A emitting green light. For example, orthographic projections of the first light-emitting elements **11**A of the first display area **101** on the substrate surface S of the base substrate **100** (for example, the upper surface of the base substrate **100** shown in FIG. **7**) are at least partially overlapped with the plurality of leads ST**1** (as shown in FIG. **3**A, which will be described in detail later). For example, the orthographic projections of the first light-emitting elements **11**A on the substrate surface S of the base substrate **100**, as a whole, fall into the first display area **101**.
(26) For example, as shown in FIGS. **2** and **3**A, the display substrate **1** further includes a plurality of first pixel driving circuits **14**A. Orthographic projections of the plurality of first pixel driving circuits **14**A on the substrate surface S of the base substrate **100** are located in the second display area **102**. The plurality of first pixel driving circuits **14**A are electrically connected with the plurality of first light-emitting elements **11**A (as shown in FIG. **4**), for example, in one-to-one correspondence, and are configured to drive the plurality of first light-emitting elements **11**A to emit light, respectively. For example, in FIG. **2** and FIG. **3**A, the plurality of rectangular dashed boxes arranged in an array each represent the position of one pixel driving circuit.
(27) It should be noted that, the circuit structure, the wiring layout and the like shown in the drawings of the embodiments of the present disclosure are all schematic, and the embodiments of the present disclosure are not limited thereto.
(28) FIG. **5**A is a partial structural diagram of a pixel driving circuit of a display substrate provided by at least one embodiment of the present disclosure. FIG. **5**A is a schematic diagram showing the first pixel driving circuit **14**A. FIG. **5**B is a partial structural diagram of a pixel driving circuit and a light-emitting element of a display substrate provided by at least one embodiment of the present disclosure. FIG. **5**B shows a partial structure of the first display area **101**.
(29) For example, as shown in FIG. **5**A, the display substrate **1** further includes a plurality of connecting traces LS. Each of the plurality of connecting traces LS is correspondingly connected to one pixel driving circuit. For example, each rectangular box in the figure (there are two intersecting lines in each rectangular box) represents the position of one pixel driving circuit. For example, a plurality of first pixel driving circuits **14**A is provided in the lowermost row in FIG. **5**A. It should be noted that, the first pixel driving circuits **14**A and the first light-emitting elements **11**A are arranged, for example, in one to one correspondence.
(30) For example, as shown in FIG. **5**A and FIG. **5**B, among the plurality of connecting traces LC, one end (the lower end in the figure) of at least one connecting trace LS, such as the connecting trace LS**1** in the figure, is electrically connected with at least one first light-emitting element **11**A. The other end (the upper end in the figure) of the at least one connecting trace, such as the connecting trace LS**1** in the figure, is electrically connected to the first pixel driving circuit **14**A. For example, as shown in the figure, the four first pixel driving circuits **14**A in the lowermost row

are each connected to one of the first light-emitting element **1101**A, the first light-emitting element **1102**A, the first light-emitting element **1103**A and the first light-emitting element **1104**A through a connecting trace LS**1**.

(31) For example, as shown in FIGS. **2** and **3**A, the display substrate **1** further includes a plurality of data lines DAT. The plurality of data lines DAT are located in the second display area **102** and extend along the first direction Y. The plurality of data lines DAT are configured to supply data signals to the plurality of first pixel driving circuits **14**A. For example, each of the plurality of data lines DAT provides a data signal to one column of pixel driving circuits of the display substrate **1** arranged along the second direction X and extending along the first direction Y.

(32) For example, as shown in FIGS. **2** and **3**A, the display substrate **1** further includes a plurality of leads ST**1**. At least some of the plurality of leads are located in the first display area **101**, and are configured to provide data signals to the plurality of first pixel driving circuits **14**A. For example, the plurality of leads ST**1** are electrically connected to the plurality of data lines DAT in one-to-one correspondence (for example, the leads ST**1** are connected to the data lines DAT at the solid dots in the figure), and are also electrically connected to the contact pad BD**1** of the bonding area in order to provide the data signals provided by the signal input elements to the first pixel driving circuits. Orthographic projections of at least some of the plurality of leads ST**1** on the substrate surface S of the base substrate **100** are overlapped with the first display area **101**.

(33) For example, as shown in FIG. **4**, the orthographic projection of the first light-emitting element **11**A on the substrate surface S of the base substrate **100** is at least partially overlapped with the orthographic projections of the plurality of leads ST**1** on the substrate surface S of the base substrate **100**. That is, in the direction perpendicular to the substrate surface S of the base substrate **100**, the plurality of leads ST**1** are routed at the side of the first light-emitting element **11**A close to the base substrate **100**. According to the embodiment of the present disclosure, the plurality of leads ST**1** are moved upwardly towards a direction close to the second display area **102** along the first direction Y, and at least some of the plurality of leads ST**1** are overlapped with the first display area **101**, so that the at least some of the plurality of leads ST**1** are located at the side of the first light-emitting element **11**A close to the base substrate **100** in the direction perpendicular to the substrate surface S of the base substrate **100**; in this way, the lower bezel of the display substrate **1** is narrowed.

(34) It should be noted that, in the embodiment of the present disclosure, the display area **10** includes a first display area **101** close to the contact pad BD**1** in the first direction Y and a second display area **102** away from the contact pad BD**1** in the first direction Y.

(35) For example, in the present disclosure, the included angle between the first direction Y and the second direction X is between 70° and 90°, inclusively. For example, the included angle between the first direction Y and the second direction X is 70°, 90° or 80° and the like, which can be set according to actual situations without limited by the embodiment of the present disclosure. For example, the included angle between the first direction Y and the second direction X can also be 75°, 85°, etc.

(36) For example, the base substrate **100** may be a glass substrate, a quartz substrate, a metal substrate, a resin substrate or the like. For example, the material of the base substrate can include organic materials, such as polyimide, polycarbonate, polyacrylate, polyetherimide, polyethersulfone, polyethylene terephthalate, polyethylene naphthalate, and other resin materials; for example, the base substrate **100** can be a flexible substrate or a non-flexible substrate, which are not limited by the embodiments of the present disclosure.

(37) For example, in some embodiments, as shown in FIGS. **3**A and **4**, the orthographic projections of at least some of the plurality of first light-emitting elements **11**A on the substrate surface S of the base substrate **100** have no overlap with the orthographic projections of the plurality of first pixel driving circuits **14**A and the plurality of data lines DAT on the substrate surface S of the base substrate **100**. For example, in FIG. **4**, the orthographic projections of the plurality of first light-

emitting elements **11**A on the substrate surface S of the base substrate **100** have no overlap with the orthographic projections of the first pixel driving circuit **14**A and the data line DAT on the substrate surface S of the base substrate **100**. For example, in FIG. **5**B, the orthographic projections, on the substrate surface S of the base substrate **100**, of the first light-emitting element **1101**A, the first light-emitting element **1102**A, the first light-emitting element **1103**A and the first light-emitting element **1104**A located in the lowermost row have no overlap with the orthographic projections of the first pixel driving circuit **14**A and the data line DAT on the substrate surface S of the base substrate **100**. In this way, the plurality of leads ST**1** are moved upwardly in a direction towards the second display area **102** along the first direction Y, and at least some (or all) of the plurality of leads ST**1** are overlapped with the first display area **101**, so as to facilitate narrowing the lower bezel of the display substrate **1**.

(38) For example, in some embodiments, as shown in FIGS. **2** and **3**A, the display substrate **1** further includes a plurality of second pixel driving circuits **14**B. The plurality of second pixel driving circuits **14**B are located at a side of the plurality of first pixel driving circuits **14**A away from the contact pad BD**1** in the first direction Y.

(39) FIG. **6**A is another partial structural diagram of a pixel driving circuit of a display substrate provided by at least one embodiment of the present disclosure. FIG. **6**A shows a schematic diagram of the pixel driving circuit in the second display area. FIG. **6**B is another partial structural diagram of a pixel driving circuit and a light-emitting element of a display substrate provided by at least one embodiment of the present disclosure. FIGS. **6**A and **6**B illustrate a partial structure of the second display area **102**.

(40) For example, in some embodiments, as shown in FIGS. **6**A and **6**B, the display substrate **1** further includes a plurality of second light-emitting elements **11**B. The plurality of second pixel driving circuits **14**B and the plurality of second light-emitting elements **11**B are located in the second display area **102**. The plurality of second pixel driving circuits **14**B are configured to drive the plurality of second light-emitting elements **11**B to emit light. As shown in FIG. **2**, the plurality of second pixel driving circuits **14**B are electrically connected to a plurality of data lines DAT. For example, the second pixel driving circuits **14**B located in the same column are electrically connected to the same data line DAT. The plurality of data lines DAT are configured to provide data signals to the plurality of second pixel driving circuits **14**B. Orthographic projections of the plurality of second light-emitting elements **11**B on the substrate surface S of the base substrate **100** are at least partly overlapped with orthographic projections of the plurality of second pixel driving circuits **14**B on the substrate surface S of the base substrate **100**, for example, they are partially overlapped.

(41) It should be noted that in the embodiment of the present disclosure, the orthographic projection of the pixel driving circuit on the substrate surface S of the base substrate **100** is shown in the figures as a rectangular area which represents the area where the pixel driving circuit is located, but the embodiments of the present disclosure are not limited thereto.

(42) For example, in some embodiments, as shown in FIGS. **6**A and **6**B, the plurality of second pixel driving circuits **14**B include a plurality of second pixel circuit groups **14**BZ arranged along the first direction Y and extending along the second direction X. Each of the second pixel circuit groups **14**BZ includes a plurality of rows of second pixel driving circuits **14**B extending along the second direction X. For example, as shown in the figures, the case where each of the second pixel circuit groups **14**BZ includes eight second pixel driving circuits **14**B arranged in two rows and four columns is described. The plurality of second pixel driving circuits **14**B of the second pixel circuit group **14**BZ are arranged in four columns along the first direction Y and arranged in two rows along the second direction X.

(43) For example, as shown in FIGS. **5**A and **5**B, the plurality of first pixel driving circuits **14**A include a plurality of first pixel circuit groups **14**AZ arranged along the first direction Y and extending along the second direction X. Each of the first pixel circuit groups **14**AZ includes a

plurality of rows of first pixel driving circuits **14**A extending along the second direction X. For example, as shown in the figures, the case where each of the first pixel circuit groups **14**AZ includes eight first pixel driving circuits **14**A arranged in two rows and four columns is described. The plurality of first pixel driving circuits **14**A of the first pixel circuit group **14**AZ are arranged in four columns along the first direction Y and arranged in two rows along the second direction X.
(44) For example, as shown in FIG. **2**, the plurality of rows of second pixel driving circuits **14**B are located at a side of the plurality of rows of first pixel driving circuits **14**A away from the contact pad BD**1** in the first direction Y. That is, the plurality of rows of second pixel driving circuits **14**B are located at a side of the plurality of rows of first pixel driving circuits **14**A away from the plurality of leads ST**1**.
(45) FIG. **5**C is a schematic diagram of a pixel circuit group of a display substrate before and after compressed as provided by at least one embodiment of the present disclosure.
(46) For example, in some embodiments, as shown in FIG. **3**B, a size Y**11** of the first pixel driving circuit **14**A in the first direction Y is as same as a size Y**12** of the second pixel driving circuit **14**B in the first direction Y. That is, the pixel driving circuits of the display substrate **1** are compressed, as a whole, in the first direction Y. At this time, all the pixel driving circuits still have the same size, which can effectively avoid the risk of abnormal display caused by different driving circuits. As shown in FIG. **5**C, taking the second pixel driving circuit **14**A as an example. The plurality of second pixel driving circuits **14**A are compressed in the first direction Y. For example, the size of two rows of second pixel driving circuits **14**A in the first direction Y before compressed is PZ**30**; the size of two rows of second pixel driving circuits **14**A in the first direction Y after compressed is PZ**3**; and the size PZ**3** is smaller than the size PZ**30**. For example, when the pixel driving circuit is compressed in the first direction Y, it can also be compressed in the second direction X at the same time. For example, the size of two rows of second pixel driving circuits **14**A in the second direction X before compressed is H**10**; the size of two rows of second pixel driving circuits **14**A in the second direction X after compressed is H**1**; and the size H**1** is smaller than the size H**10**. The compression in the second direction X can reduce the length of the connecting trace LS, which is beneficial for the stability of the electrical signals of the light-emitting elements.
(47) For example, in the embodiment of the present disclosure, the pixel driving circuit (for example, the first pixel driving circuit **14**A or the second pixel driving circuit **14**B) includes conventional 2T1C (i.e., two transistors and one capacitor) pixel circuit, 7T1C (i.e., seven transistors and one capacitor) pixel circuit or the like. The pixel driving circuit **14** includes at least one switching transistor and one first transistor (such as the first transistor **12** in FIG. **7**); a gate electrode of the switching transistor receives a gate scanning signal, and a source electrode or a drain electrode of the switching transistor is connected to the data line DAT to receive a data signal. In different embodiments, the pixel driving circuit may further include a compensation circuit, which includes an internal compensation circuit or an external compensation circuit; and the compensation circuit may include transistors, capacitors and the like. For example, the pixel circuit may further include a reset circuit, a light-emitting control circuit, a detection circuit or the like, depending on demands. The embodiment of the present disclosure is not intended to limit the type of the first light-emitting element and the specific structure of the pixel circuit.
(48) For example, in the embodiment of the present disclosure, the compression of the pixel driving circuit can be realized in various ways. For example, by narrowing the gap between the traces in the second direction X or the first direction Y under the condition that the line width or layout of traces of the pixel driving circuit remains unchanged. For example, under the condition that the gap between the traces of the pixel driving circuit remains unchanged, it can reduce the line width or length of the traces. The present disclosure is not intended to limit the compression process of the pixel driving circuit.
(49) It should be noted that, in the embodiment of the present disclosure, the size of each pixel driving circuit is periodic. The first pixel driving circuits **14**A electrically connected with the first

light-emitting elements **11**A in the first display area have the same size, and the second pixel driving circuits **14**B electrically connected with the second light-emitting elements **11**B in the second display area have the same size.

(50) For example, in some embodiments, as shown in FIG. **3**A, the size Y**11** of the first pixel driving circuit **14**A in the first direction Y is as same as the size Y**12** of the second pixel driving circuit **14**B in the first direction Y, with a value range of about 63.8-63.9 microns. For example, when the pixel driving circuits of the display substrate **1** are compressed as a whole, reference is made to the case where the size of one pixel driving circuit in the first direction Y before compressed is about 64 microns. In such case, the lower bezel of the display substrate **1** has to be narrowed by an amount of M**1**. For example, the value of M**1** can be set according to the design requirements, for example, about 10-12 microns. For example, if the number of rows of all the pixel driving circuits of the display substrate is N**1**, the size of each row of pixel driving circuits has to be compressed by an amount which is obtained through dividing M**1** by N**1**. As a result, the value obtained by 64-M**1**/N**1** is the size of the pixel driving circuit in the first direction Y after compressed. It should be noted that, the value ranges of the size Y**11** and the size Y**12** as mentioned above are illustrated by way of example, and the size of the pixel driving circuit in the longitudinal direction after compressed can be flexibly determined based on the compressed amount of the display substrate and the number of rows of the pixel driving circuits of the display substrate, without particularly limited in the embodiments of the present disclosure.

(51) It should be noted that, the word “about” means that the value can be fluctuated within its range, such as “±5%” or “±25%”.

(52) For example, as shown in FIGS. **6**A and **6**B, the display substrate **1** further includes at least one row of first dummy pixel driving circuits **15**A extending along the second direction X. In the first direction Y, each row of the first dummy pixel driving circuits **15**A are distributed, at intervals, between the at least one row of second pixel driving circuits **14**B extending along the second direction X. For example, the pixel driving circuits of the display substrate **1** are compressed as a whole.

(53) For example, in some embodiments, as shown in FIGS. **6**A and **6**B, in the first direction Y, adjacent two rows of first dummy pixel driving circuits **15**A extending along the second direction X are spaced apart by at least one second pixel circuit group **14**BZ there-between. For example, in the first direction Y, one row of first dummy pixel driving circuits **15**A is inserted between two second pixel circuit groups **14**BZ. For example, as shown in FIG. **6**B, in the first direction Y, one row of first dummy pixel driving circuits **15**A is inserted between four rows of second pixel driving circuits **14**B. In such case, the area in the figure where the plurality of second light-emitting elements **11**B corresponding to one second pixel circuit group **14**BZ are located is labeled as **11**BZ, that is, the rectangular box with bold lines in the figure. The size PZ**10** of the area **11**BZ in the first direction Y is as same as the size PZ**20** of two second pixel circuit groups **14**BZ and one row of first dummy pixel driving circuits **15**A in the first direction Y. That is, FIG. **6**B shows one repeating unit of the second display area **102**. The first dummy pixel driving circuit **15**A and the second pixel driving circuit **14**B have the same circuit design, so as to maintain the uniformity of wirings and improve the display uniformity of the second display area **102**. The first dummy pixel driving circuit **15**A is not connected to the second light-emitting element **11**B.

(54) For example, as shown in FIGS. **5**A and **5**B, in the first direction Y, the first dummy pixel driving circuit **15**A is not provided between the plurality of rows of first pixel driving circuits **14**A extending in the second direction X. In this way, the size in the first direction Y of the area where the first pixel circuit group **14**AZ composed of a plurality of rows of first pixel driving circuits **14**A is located is smaller than the size in the first direction Y of the area where the plurality of first light-emitting elements **11**A corresponding to the first pixel circuit group **14**AZ are located, so that the first light-emitting elements **11**A and the first pixel driving circuits **14**A are arranged in a staggered manner. For example, as shown in FIGS. **2** and **3**A, the edge BY**2** of the first pixel driving circuit

14A that is parallel to the second direction X and closest to the bonding area **21** is closer to the second display area **102** than the edge BY**1** (as shown in FIG. **3**A) of the display area **10** that is parallel to the second direction X and closest to the bonding area **21**. In this way, the plurality of leads ST**1** are moved upwardly in a direction close to the second display area **102** along the first direction Y, and at least some of the plurality of leads ST**1** are overlapped with the first display area **101**, for example, all of the plurality of leads ST**1** are overlapped with the first display area **101**, so as to facilitate narrowing the lower bezel of the display substrate **1**.

(55) For example, in some embodiments, as shown in FIGS. **5**A, **5**B, **6**A and **6**B, the display substrate **1** further includes at least one column of second dummy pixel driving circuits **15**B extending along the first direction Y. For example, one column of second dummy pixel driving circuits **15**B is provided between four columns of pixel driving circuits. In this way, it can improve the uniformity of traces of the pixel driving circuits of the display substrate **1** so as to improve the display uniformity.

(56) It should be noted that, in the embodiment of the present disclosure, the dummy pixel driving circuit and the pixel driving circuit have the same structure, but the dummy pixel driving circuit is not electrically connected with the light-emitting element.

(57) FIG. **3**B is a schematic diagram of yet another display substrate provided by at least one embodiment of the present disclosure.

(58) For example, in some embodiments, as shown in FIG. **3**B, the size Y**12** of the second pixel driving circuit **14**B in the first direction Y is greater than the size Y**11** of the first pixel driving circuit **14**A in the first direction Y. In such case, the pixel driving circuits of the display substrate **1** are partially compressed. For example, several rows of pixel driving circuits on the display substrate **1** close to the peripheral area **20** are compressed, so that the plurality of leads are moved upwardly towards the second display area **102** along the first direction Y, and at least some of the plurality of leads are overlapped with the first display area **101**; in this way, the pixel driving circuits and the light-emitting elements are arranged in a staggered manner to narrow the lower bezel.

(59) For example, in some embodiments, as shown in FIG. **3**B, the size Y**12** of the second pixel driving circuit **14**B in the first direction Y is in a range of about 63-65 microns, and the size Y**11** of the first pixel driving circuit **14**A in the first direction Y is in a range of about 60-62 microns. For example, when the pixel driving circuits on the display substrate **1** are partially compressed, the size Y**12** of the second pixel driving circuit **14**B in the first direction Y is uncompressed. For example, in such case, the size of the lower bezel of the display substrate **1** needs to be compressed by an amount of M**1**. For example, the value of M**1** can be set according to the design requirements, for example, 10-12 microns. For example, if the number of the rows of the pixel driving circuits of the display substrate that need to be compressed is X**1**, the amount of the size of each row of pixel driving circuits that needs to be compressed can be calculated by dividing M**1** by X**1**. As a result, the value obtained by Y**11**-M**1**/X**1** is the size Y**11** of the pixel driving circuit in the first direction Y after compressed. It should be noted that, the value ranges of the size Y**11** and the size Y**12** as mentioned above are illustrated by way of example, and the size of the pixel driving circuit in the longitudinal direction after compressed can be flexibly determined based on the compressed amount of the display substrate and the number of the rows of the pixel driving circuits of the display substrate that need to be compressed, without particularly limited in the embodiments of the present disclosure.

(60) For example, in some embodiments, as shown in FIGS. **5**A, **5**B, **6**A and **6**B, the plurality of connecting traces LS include a first connecting trace LS**1** partially located in the first display area **101** and a second connecting trace LS**2** located in the second display area; the second connecting trace LS**2** electrically connects the second pixel driving circuit **14**B and the second light-emitting element **11**B; and the first connecting trace LS**1** electrically connects the first pixel driving circuit **14**A and the first light-emitting element **11**A. The light emitted by the second light-emitting

element **11**B that is electrically connected with the second connection line LS**2** and the light emitted by the first light-emitting element **11**A that is electrically connected with the first connection line LS**1** have the same color, for example, blue light. The line width of the second connecting trace LS**2** is greater than that of the first connecting trace. In such case, because the first pixel driving circuit **14**A and the first light-emitting element **11**A are arranged in a staggered manner, the length of the first connecting trace LS**1** at the side close to the contact pad BD**1** is greater than the length of the second connecting trace LS**2**. With the increase of the length of the traces, the line width of the first connecting trace LS**1** decreases due to restrict of the arrangement space. By providing the second connecting trace LS**2** with a larger line width, it can reduce the parasitic capacitance and voltage drop on the connecting trace and ensure the stability of the display performance of the display substrate **1** as far as possible.
(61) It should be noted that, in the embodiment of the present disclosure, the line width refers to the width of the connecting trace in the direction perpendicular to its wiring direction.
(62) For example, in some embodiments, as shown in FIGS. **2** and **3**A, the peripheral area **20** further includes a bending area **23**. The bending area **23** is located at a side of the first display area **101** away from the second display area **102**. For example, the bending area **23** is located between the bonding area **21** and the first display area **101**. The bending area **23** includes a plurality of signal lines **231** extending along the first direction Y and connected with a plurality of leads ST**1** in one-to-one correspondence. The plurality of signal lines **231** are arranged at intervals in the second direction X. The plurality of signal lines **231** are also connected to the contact pads BD**1** of the bonding area **21** in one-to-one correspondence so as to transmit electrical signals.
(63) For example, in some embodiments, as shown in FIG. **3**A, the plurality of leads ST**1** include a plurality of first leads ST**11** and a plurality of second leads ST**12**. The plurality of first leads ST**11** are connected in one-to-one correspondence with the plurality of second leads ST**12**. The plurality of first leads ST**11** are located in the first display area **101**, and the plurality of second leads ST**12** are located in the peripheral area **20** and are connected in one-to-one correspondence with the plurality of signal lines **231**. The first lead ST**11** is closer to the second display area **102** than the second lead ST**12**. As shown in FIG. **2**, the plurality of data lines DAT have a first average line spacing PD**1** in the second direction X. As shown in FIG. **3**A, the plurality of signal lines **231** have a second average line spacing PD**7** in the second direction X, and the plurality of first leads ST**11** have a third average line spacing PD**5** in the second direction X. The third average line spacing PD**5** is between the first average line spacing PD**1** and the second average line spacing PD**7**. At this time, the third average line spacing PD**5** of the plurality of leads ST**1** is gradually narrowed in the first display area **101**, the plurality of leads ST**1** are moved upwardly towards the second display area **102** along the first direction Y, and at least some of the plurality of leads ST**1** are overlapped with the first display area **101**, for example, the plurality of leads ST**1** are located in the first display area **101**, so as to reduce the lower bezel of the display substrate.
(64) For example, in some embodiments, as shown in FIG. **3**A, the third average line spacing PD**5** between the first leads ST**11** decreases in the first direction Y from a side close to the plurality of data lines DAT to another side away from the plurality of data lines DAT. That is, the third average line spacing PD**5** of the plurality of first leads ST**11** is gradually narrowed from top to bottom, so that the plurality of first leads ST**11** form a gradually narrowed fan shape.
(65) It should be noted that, in the embodiment of the present disclosure, the average line spacing refers to an average value of the spacing distances between the plurality of traces along the second direction X.
(66) For example, in some embodiments, as shown in FIGS. **2** and **3**A, the display area **10** includes a light-emitting layer LC**1** and a pixel circuit layer DR**1** which are stacked, the light-emitting layer LC**1** is located at the side of the pixel circuit layer DR**1** away from the base substrate **100**, and the display area **10** also includes a plurality of pixel unit groups arranged in rows and columns in an array, each of the plurality of pixel unit groups includes a plurality of sub-pixels arranged in at least

one row, and each of the plurality of sub-pixels includes a pixel driving circuit (including a first pixel driving circuit **14**A and a second pixel driving circuit **14**B) and a light-emitting element (including a first light-emitting element **11**A and a second light-emitting element **11**B). The pixel driving circuit is configured to drive the light-emitting element to emit light. The light-emitting layer LC**1** has a first edge BY**1** which is opposite to the bending region **23** and parallel to the second direction X. The plurality of leads ST**1** have a fifth average line spacing PD**3** at a position overlapped with the first edge BY**1** of the light-emitting layer LC**1**. The first average line spacing PD**1** is larger than the second average line spacing PD**7**, and the fifth average line spacing PD**3** is between the first average line spacing PD**1** and the second line spacing PD**7**. At this time, at least some of the plurality of leads ST**1** are located in the first display area **101**, thereby reducing the lower bezel of the display substrate **1**.
(67) For example, in some embodiments, as shown in FIGS. **2** and **3**A, the pixel circuit layer DR**1** has a second edge BY**2** opposite to the bending region **23** and parallel to the second direction X, and the first edge BY**1** of the light-emitting layer LC**1** (shown in FIG. **3**A) is closer to the bending region **23** than the second edge BY**2** of the pixel circuit layer DR**1**.
(68) For example, in some embodiments, as shown in FIG. **3**A, a plurality of signal lines **231** or a plurality of leads ST**1** of the bending region **23** intersects with the first edge BY**1** in a direction perpendicular to the base substrate **100**. For example, the plurality of leads ST**1** are overlapped with the first edge BY**1** of the light-emitting layer LC**1**, or the plurality of signal lines **231** are overlapped with the first edge BY**1** of the light-emitting layer LC**1**, so that the positions of the plurality of leads ST**1** are moved upwardly relative to the light-emitting layer LC**1**, and at least some of the plurality of leads ST**1** are located in the orthographic projection of the light-emitting layer LC**1** on the substrate surface S of the base substrate **100**, thereby reducing the lower bezel of the display substrate **1**.
(69) For example, in some embodiments, as shown in FIG. **2** and FIG. **3**A, when connection points (shown by solid dots in the figure) of the plurality of leads ST**1** and the plurality of data lines DAT are located in the second display area **102**, in the case where the plurality of leads ST**1** intersect with the second edge BY**2** in the direction perpendicular to the base substrate **100**, the plurality of leads ST**1** have a sixth average line spacing PD**4** (shown in FIG. **2**) at the position of the second edge BY**2**, and the sixth average line spacing PD**4** is between the first average line spacing PD**1** and the third average line spacing PD**3**. At this time, portions of the plurality of leads ST**1** are overlapped with portions of the light-emitting layer LC**1** located in the first display area **101**. For example, in some embodiments, as shown in FIG. **5**B, the light-emitting elements (including the first light-emitting elements **11**A and the second light-emitting elements **11**B) of sub-pixels of a plurality of pixel unit groups arranged in rows and columns in an array have a first average column period length PZ**1** in the first direction Y. As shown in FIG. **3**A, the pixel driving circuits (including the first pixel driving circuits **14**A and the second pixel driving circuits **14**B) of sub-pixels of a plurality of pixel unit groups arranged in rows and columns in an array have a second average column period length Y**12** in the first direction Y; the second average column period length Y**12** is smaller than the first average column period length PZ**1**. Therefore, at the side of the display area close to the bending area **23**, the light-emitting elements and the pixel driving circuits are arranged in a staggered manner to reduce the size of the lower frame.
(70) For example, as shown in FIGS. **6**A and **6**B, in the case where the first dummy pixel driving circuit **15**A is not provided, the pixel driving circuits of the sub-pixels of a plurality of pixel unit groups arranged in rows and columns in an array have a column period length PZ**2** in the second display area **102** of the display area **10** in the first direction Y. For example, the column period length PZ**2** is equal to the first average column period length PZ**1** (shown in FIG. **6**B). The first average column period length PZ**1** is the size, in the first direction Y, of an area where the plurality of light-emitting elements connected to the second pixel circuit group **14**BZ are located. At this time, the pixel driving circuits of the display substrate **1** are compressed, as a whole, in the first

direction Y.
(71) For example, as shown in FIG. **5**A, in the case where the pixel driving circuits of the display substrate **1** are compressed as a whole in the first direction Y, one row of first pixel driving circuit groups **14**AZ of the display area **10** closest to the bending area **23** has a column period length PZ**3**. For example, the column period length PZ**3** is smaller than the first average column period length PZ**1**. At this time, the size, in the first direction Y, of the plurality of rows of first pixel driving circuits **14**A of the display area **10** that are close to the bending area **23** is compressed, so that the plurality of leads are moved upwardly in a direction close to the second display area along the first direction, and at least some of the plurality of leads are overlapped with the first display area to reduce the lower frame of the display substrate **1**.
(72) For example, in some embodiments, as shown in FIG. **3**A, the spacing between one row of first pixel driving circuits **14**A closest to the bending region **23** and the first edge BY**1** of the light-emitting layer LC**1** is JD**3** (along the first direction Y). For example, the spacing JD**3** may also indicate the size of a portion of the light-emitting layer LC**1** that extends beyond the first pixel driving circuit **14**A in the first direction Y. For example, the spacing JD**3** may be greater than or equal to the size PZ**3** of two rows of second pixel driving circuits **14**A in the first direction Y. For example, the size of the spacing JD**3** can be designed according to the size by which the lower frame needs to be narrowed, but the embodiments of the present disclosure are not limited thereto.
(73) FIG. **7** is a schematic cross-sectional view of a display substrate provided by at least one embodiment of the present disclosure. For example, FIG. **7** is a schematic cross-sectional view of one sub-pixel of the first display area **10**, and the cross-sectional views of other sub-pixels can refer to FIG. **7**, which will not be repeated here.
(74) For example, in some embodiments, as shown in FIG. **7**, the display substrate **1** further includes a first metal layer **301**, a second metal layer **302** and a third metal layer **303**. The first metal layer **301** is located on the base substrate **100**, the second metal layer **302** is located at a side of the first metal layer **301** away from the base substrate **100**, and the third metal layer **303** is located at a side of the second metal layer **302** away from the base substrate **100**. As shown in FIG. **4**, a plurality of leads ST**1** is arranged in the same layer as the first metal layer **301** or the second metal layer **302** to simplify the manufacturing process.
(75) For example, in some embodiments, as shown in FIG. **4** and FIG. **7**, some of the plurality of leads ST**1** (for example, the lead ST**21**) are arranged in the same layer as the first metal layer **301**, and some other of the plurality of leads ST**1** (for example, the lead ST**22**) are arranged in the same layer as the second metal layer **302**; and the leads ST**1** located in different layers are alternately arranged. For example, the leads ST**21** and ST**22** are located in different layers and are alternately arranged to reduce the wiring space and the crosstalk between signals.
(76) It should be noted that, in the embodiment of the present disclosure, “arranged in the same layer” includes the case where two functional layers or structural layers are in the same layer of the display substrate in terms of hierarchical structure and are formed of the same material; that is, in the manufacturing process, the two functional layers or structural layers can be formed from the same material layer, and the required patterns and structures can be formed by the same patterning process. A single patterning process includes, for example, steps of forming a photoresist, exposing the photoresist, developing the photoresist, etching the photoresist and others.
(77) For example, the materials of the first metal layer **301**, the second metal layer **302** and the third metal layer **303** may include metal materials or alloy materials, such as a metallic single-layered or multi-layered structure formed of molybdenum, aluminum, titanium, etc. For example, the multi-layered structure is a multi-metal lamination (such as a three-layered metal lamination of titanium, aluminum and titanium (Ti/Al/Ti)). For example, the materials of the first metal layer **301**, the second metal layer **302** and the third metal layer **303** may be the same or different, and the embodiments of the present disclosure are not limited thereto.
(78) For example, in some embodiments, as shown in FIG. **7**, the display substrate **1** further

includes a first insulating layer **1245** (e.g., the first planarization layer), a second insulating layer **1247** (e.g., the second planarization layer) and a third insulating layer **1248** (e.g., the third planarization layer) which are located at a side of the plurality of first pixel driving circuits **14**A away from the base substrate **100**. For example, the second insulating layer **1247** is located between the first insulating layer **1245** and the third insulating layer **1248**. The connecting trace LS**1** is arranged at a side of the second insulating layer **1247** away from the base substrate **100**.
(79) As shown in FIG. **5**A, FIG. **5**B and FIG. **7**, one end of the connecting trace LS**1** is electrically connected with the corresponding first light-emitting element **11**A through a via hole VH**2** in the third insulating layer **1248**.
(80) For example, the materials of the first insulating layer **1245**, the second insulating layer **1247** and the third insulating layer **1248** include inorganic insulating materials such as silicon oxide, silicon nitride, and silicon oxynitride, and may also include organic insulating materials such as polyimide, polyamide, acrylic resin, benzocyclobutene, or phenolic resin, without limited by the embodiments of the present disclosure.
(81) For example, as shown in FIG. **7**, the display substrate **1** further includes a transition electrode TS**1** connected to a first pixel driving circuit **14**A or a second pixel driving circuit **14**B (not shown in FIG. **7**). The transition electrode TS**1** is located at a side of the first insulating layer **1245** away from the base substrate **100**. The transition electrode TS**1** is electrically connected to a drain electrode **124** of a first transistor **12** in the corresponding first pixel driving circuit **14**A through a via hole **252** in the first insulating layer **1245**, and is also connected to a corresponding connecting trace LS**1** through a via hole VH**1** in the second insulating layer **1247**.
(82) For example, as shown in FIG. **7**, the display substrate further includes a first gate insulating layer **1242**, a second gate insulating layer **1243**, an interlayer insulating layer **1244** and an active layer **121**. The interlayer insulating layer **1244** is arranged at a side of the third metal layer **303** close to the base substrate **100**. The second gate insulating layer **1243** is arranged at a side of the interlayer insulating layer **1244** close to the base substrate **100**. The first gate insulating layer **1242** is arranged at a side of the second gate insulating layer **1243** close to the base substrate **100**. The second metal layer **302** is located between the interlayer insulating layer **1244** and the second gate insulating layer **1243**. The first metal layer **301** is located between the second gate insulating layer **1243** and the first gate insulating layer **1242**. The active layer **121** is located at a side of the first gate insulating layer **1242** close to the base substrate **100**.
(83) For example, the material of one or more of the first gate insulating layer **1242**, the second gate insulating layer **1243** and the interlayer insulating layer **1244** may include an insulating material such as silicon oxide, silicon nitride and silicon oxynitride. The materials of the first gate insulating layer **1242**, the second gate insulating layer **1243** and the interlayer insulating layer **1244** may be the same or different.
(84) For example, the material of the active layer **121** may include polysilicon or oxide semiconductor (for example, indium gallium zinc oxide (IGZO)).
(85) For example, as shown in FIG. **7**, the display substrate **1** may further include a buffer layer **1241** and a barrier layer **1240**. The buffer layer **1241** is located at a side of the active layer **121** close to the base substrate **100**, and the barrier layer **1240** is located at a side of the buffer layer **1241** close to the base substrate **100**. The buffer layer **1241** serves as a transition layer, which can not only prevent harmful substances in the base substrate from invading into the display substrate, but also increase the adhesion of the film layers of the display substrate **1** onto the base substrate **100**. The barrier layer **1240** can provide a flat surface for forming the first pixel driving circuit **14**A, and can prevent impurities possibly existed in the base substrate **100** from diffusing into the first pixel driving circuit **14**A and adversely affecting the performance of the display substrate **1**.
(86) For example, as shown in FIG. **7**, the first pixel driving circuit **14**A or the second pixel driving circuit **14**B further includes a first transistor **12** and a storage capacitor **13**. The first transistor **12** includes a transistor directly electrically connected to the first light-emitting element **11**A, such as a

switching transistor (e.g., a light-emitting control transistor) or a driving transistor. The first transistor **12** includes a gate electrode **122**, a source electrode **123**, a drain electrode **124** and an active layer **121**; and the storage capacitor **13** includes a first electrode plate **131** and a second electrode plate **132**. The gate electrode **122** is located in the first metal layer **301**, and the source-drain electrodes (source electrode **123** and drain electrode **124**) are located in the third metal layer **303**. For example, the first electrode plate **131** is located in the first metal layer **301** and the second electrode plate **132** is located in the second metal layer **302**. The gate electrode **122** and the first electrode plate **131** are arranged in the same layer. The first electrode plate **131** and the second electrode plate **132** are spaced apart by the second insulating layer **1243**, so as to realize a capacitor function.
(87) For example, in other embodiments, the first electrode plate **131** may be located in the second metal layer **302**, and the second electrode plate **132** may be located in the third metal layer **303**. At this time, the first electrode plate **131** and the second electrode plate **132** are spaced apart by the interlayer insulating layer **1244**. The embodiments of the present disclosure are not limited to the specific arrangement of the storage capacitor **13**.
(88) For example, as shown in FIG. **7**, the display substrate **1** further includes a passivation layer **1246**, which is located at a side of the first insulating layer **1245** close to the base substrate **100**. At this time, the via hole **252** also penetrates through the passivation layer **1246**. The passivation layer **1246** can protect the source electrode **123** and the drain electrode **124** of the first pixel driving circuit **14**A from water vapor corrosion. The drain electrode **124** of the first transistor **12** in the first pixel driving circuit **14**A and the transition electrode TS**1** are connected through the via hole **252**.
(89) For example, the material of the passivation layer **1246** may include an organic insulating material or an inorganic insulating material, such as silicon nitride material. The silicon nitride material has high dielectric constant and good hydrophobic function, so that the first pixel driving circuit **14**A can be well protected against water vapor corrosion.
(90) For example, as shown in FIG. **7**, the display substrate **1** further includes a pixel defining layer **146**. The first light-emitting element **11**A is arranged at a side of the second insulating layer **1245** away from the base substrate **100**. The first light-emitting element **11**A includes a first electrode **113** (e.g., anode), a light-emitting layer **112** and a second electrode **111** (e.g., cathode). The first electrode **113** is located at a side of the second insulating layer **1245** away from the base substrate **100**, and is connected to the connecting trace LS**1** through the via hole VH**2**. The second electrode **111** is located at a side of the pixel defining layer **146** away from the base substrate **100**. The pixel defining layer **146** is located at a side of the first electrode **113** away from the base substrate **100** and includes a first pixel opening **1461**. The first pixel opening **1461** is arranged to be corresponding to the first light-emitting element **11**A. The light-emitting layer **112** is located in the first pixel opening **1461**, and is located between the first electrode **113** and the second electrode **111**. A part of the light-emitting layer **112** directly sandwiched between the first electrode **113** and the second electrode **111** will emit light after being energized, so that an area occupied by this part corresponds to the light-emitting area of the first light-emitting element **11**A.
(91) For example, the first pixel driving circuit **14**A generates a light-emitting driving current under the control of the data signal provided by the data line DAT, the gate scanning signal and the light-emitting control signal provided by the shift register, the power signal and the like, and the light-emitting driving current enables the first light-emitting element **11**A to emit red light, green light, blue light, or white light, etc.
(92) For example, the material of the pixel defining layer **146** may include an organic insulating material such as polyimide, polyamide, acrylic resin, benzocyclobutene or phenolic resin, or an inorganic insulating material such as silicon oxide and silicon nitride, without limited by the embodiments of the present disclosure.
(93) For example, the material of the first electrode **113** may include at least one transparent conductive oxide material, including indium tin oxide (ITO), indium zinc oxide (IZO), zinc oxide

(ZnO) and the like. Furthermore, the first electrode **113** may include a metal with high reflectivity as a reflective layer, such as silver (Ag).
(94) For example, the light-emitting layer **112** for OLED can include small molecular organic materials or polymer molecular organic materials, can be fluorescent light-emitting materials or phosphorescent light-emitting materials, and can emit red light, green light, blue light or white light; furthermore, the light-emitting layer may further include functional layers such as electron injection layer, electron transport layer, hole injection layer and hole transport layer as required.
(95) For example, the light-emitting layer **112** for QLED may include quantum dot materials, such as silicon quantum dots, germanium quantum dots, cadmium sulfide quantum dots, cadmium selenide quantum dots, cadmium telluride quantum dots, zinc selenide quantum dots, lead sulfide quantum dots, lead selenide quantum dots, indium phosphide quantum dots, indium arsenide quantum dots, etc., and the particle size of the quantum dots is 2-20 nm.
(96) For example, the second electrode **111** may include various conductive materials. For example, the second electrode **111** may include metal materials such as lithium (Li), aluminum (Al), magnesium (Mg), silver (Ag), etc.
(97) For example, as shown in FIG. **7**, the display substrate **1** further includes a spacer **148** located at a side of the pixel defining layer **146** away from the base substrate **100**, that is, the spacer **148** is located between the pixel defining layer **146** and the second electrode **111**. For example, the material of the spacer **148** may include a transparent insulating material.
(98) For example, as shown in FIG. **7**, the display substrate **1** further includes an encapsulation layer **147**. The encapsulation layer **147** is located at a side of the second electrode **111** away from the base substrate **100**. The encapsulation layer **147** seals the first light-emitting element **11**A, so that the degradation of the first light-emitting element **11**A caused by moisture and/or oxygen contained in the environment can be reduced or prevented. The encapsulation layer **147** may have a single-layered structure or a composite layer structure including a stacked structure of inorganic layers and organic layers. The encapsulation layer **147** includes at least one encapsulation sublayer. For example, the encapsulation layer **147** may include a first inorganic encapsulation layer, a first organic encapsulation layer and a second inorganic encapsulation layer which are sequentially arranged.
(99) For example, the material of the encapsulation layer **147** may include insulating materials such as silicon nitride, silicon oxide, silicon oxynitride and polymer resin. Inorganic materials such as silicon nitride, silicon oxide and silicon oxynitride have high density, which can prevent from an invasion of water and oxygen. The material of the organic encapsulation layer can be a polymer material containing a desiccant or a polymer material that can block water vapor such as a polymer resin, so as to planarize the surface of the display substrate and relieve the stress of the first inorganic encapsulation layer and the second inorganic encapsulation layer; and the material of the organic encapsulation layer can also include a water-absorbing material such as a desiccant to absorb water, oxygen and other invading substances.
(100) FIG. **8** is a schematic diagram of a display device provided by at least one embodiment of the present disclosure.
(101) At least one embodiment of the present disclosure also provides a display device. FIG. **8** is a schematic diagram of a display device provided by an embodiment of the present disclosure. As shown in FIG. **8**, the display device **2** includes the display substrate **1** provided in any embodiment of the present disclosure and signal input elements. For example, the display substrate **1** shown in FIG. **3**A or **3**B is used as the display substrate **1**.
(102) It should be noted that the display device **2** can be a wearable device. For example, the display device **2** can also be any product or component with display function such as OLED panel, OLED TV, QLED panel, QLED TV, mobile phone, tablet computer, notebook computer, digital photo frame, navigator, etc. The display device **2** may also include other components, such as a data driving circuit, a timing controller, etc., which is not limited by the embodiments of the

present disclosure.
(103) It should be noted that, for the sake of clarity and conciseness, the embodiments of the present disclosure do not show all the constituent units of the display device. In order to realize the basic functions of the display device, a person skilled in the art can provide and design other structures not shown according to specific needs, which are not limited by the embodiments of the present disclosure.
(104) For the technical effects of the display device **2** provided in the above embodiments, reference can be made to the technical effects of the display substrate **1** provided in the embodiments of the present disclosure, which will not be repeated here.
(105) The following points need to be explained: (1) The drawings of the embodiments of the present disclosure only refer to the structures related to the disclosed embodiments, and other structures can refer to the general design. (2) Without conflict, the embodiments of the present disclosure and the features in the embodiments can be combined with each other to obtain new embodiment(s).
(106) The above are only optional embodiments of the present disclosure without limiting the present disclosure. Any modifications, equivalent substitutions, improvements or the like within the spirit and principle of the present disclosure should be fallen within the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure should be delimited by the protection scope of the claims.

Claims

1. A display substrate, comprising: a base substrate, comprising a display area and a peripheral area located at least at one side of the display area, the display area comprising a first display area and a second display area, and the first display area being located between the second display area and the peripheral area in a first direction; at least one group of contact pads located in the peripheral area; a plurality of first light-emitting elements located in the first display area; a plurality of first pixel driving circuits located in the second display area; a plurality of connecting traces, one end of at least one connecting trace of the plurality of connecting traces being electrically connected with at least one first light-emitting element of the plurality of first light-emitting elements, and the other end of the at least one connecting trace being electrically connected with the first pixel driving circuit; a plurality of data lines located in the second display area and configured to provide data signals to the plurality of first pixel driving circuits; and a plurality of leads located in the first display area and in the peripheral area, the plurality of leads connecting the plurality of data lines and the at least one group of contact pads; wherein orthographic projections of the plurality of first light-emitting elements on a substrate surface of the base substrate are at least partly overlapped with orthographic projections of the plurality of leads on the substrate surface of the base substrate.
2. The display substrate according to claim 1, wherein the orthographic projections of at least some first light-emitting elements of the plurality of first light-emitting elements on the substrate surface of the base substrate have no overlap with orthographic projections of the plurality of first pixel driving circuits and the plurality of data lines on the substrate surface of the base substrate.
3. The display substrate according to claim 1, further comprising a plurality of second pixel driving circuits and a plurality of second light-emitting elements, wherein the plurality of second pixel driving circuits and the plurality of second light-emitting elements are located in the second display area, and the plurality of second pixel driving circuits are configured to drive the plurality of second light-emitting elements to emit light, the plurality of second pixel driving circuits are electrically connected with the plurality of data lines, and the plurality of data lines are configured to provide data signals to the plurality of second pixel driving circuits, and orthographic projections of the plurality of second light-emitting elements on the substrate surface of the base substrate are at least partly overlapped with orthographic projections of the plurality of second pixel driving

circuits on the substrate surface of the base substrate.
4. The display substrate according to claim 3, wherein the plurality of second pixel driving circuits comprise a plurality of second pixel circuit groups arranged along the first direction and extending along a second direction, the first direction intersects with the second direction, and each of the plurality of second pixel circuit groups comprises a plurality of rows of second pixel driving circuits extending along the second direction; the plurality of first pixel driving circuits comprise a plurality of first pixel circuit groups arranged along the first direction and extending along the second direction, and each of the plurality of first pixel circuit groups comprises a plurality of rows of first pixel driving circuits extending along the second direction; and the plurality of rows of second pixel driving circuits are located at a side of the plurality of rows of first pixel driving circuits away from the first display area.
5. The display substrate according to claim 4, further comprising at least one row of first dummy pixel driving circuits extending along the second direction, in the first direction, the at least one row of first dummy pixel driving circuits are distributed at intervals between the at least one row of second pixel driving circuits extending along the second direction.
6. The display substrate according to claim 5, wherein in the first direction, the first dummy pixel driving circuit is not provided between a plurality of rows of first pixel driving circuits extending in the second direction.
7. The display substrate according to claim 5, wherein in the first direction, two adjacent rows of first dummy pixel driving circuits extending in the second direction are spaced apart by at least one second pixel circuit group.
8. The display substrate according to claim 4, wherein a size of the second pixel driving circuit in the first direction is larger than that of the first pixel driving circuit in the first direction.
9. The display substrate according to claim 8, wherein the size of the second pixel driving circuit in the first direction is in the range from 63 microns to 65 microns, and the size of the first pixel driving circuit in the first direction is in the range from 60 microns to 62 microns.
10. The display substrate according to claim 4, wherein a size of the first pixel driving circuit and a size of the second pixel driving circuit are equal in the first direction.
11. The display substrate according to claim 10, wherein the size of the first pixel driving circuit and the size of the second pixel driving circuit are equal in the first direction, and have a value in the range from 63.8 microns to 63.9 microns.
12. The display substrate according to claim 4, wherein the plurality of connecting traces comprise a first connecting trace partially located in the first display area and a second connecting trace located in the second display area, wherein the second connecting trace is electrically connected with a corresponding second light-emitting element, the first connecting trace is electrically connected with a corresponding first light-emitting element, and the corresponding second light-emitting element electrically connected with the second connecting trace and the corresponding first light-emitting element electrically connected with the first connecting trace emit light with the same color; and a line width of the second connecting trace is larger than that of the first connecting trace.
13. The display substrate according to claim 3, further comprising a first metal layer, a second metal layer and a third metal layer, wherein the first metal layer is located on the base substrate, the second metal layer is located at a side of the first metal layer away from the base substrate, and the third metal layer is located at a side of the second metal layer away from the base substrate; the plurality of leads are arranged in the same layer as the first metal layer or the second metal layer; or some leads of the plurality of leads are arranged in the same layer as the first metal layer, and the other leads of the plurality of leads are arranged in the same layer as the second metal layer; and the some leads and the other leads are alternately arranged.
14. The display substrate according to claim 1, wherein the peripheral area further comprises a bending area which is located at a side of the first display area away from the second display area,

and the bending area comprises a plurality of signal lines, the plurality of signal lines extend along the first direction and are connected in one-to-one correspondence with the plurality of leads; wherein the first direction intersects with the second direction.
15. The display substrate according to claim 14, wherein the plurality of leads comprise a plurality of first leads and a plurality of second leads, the plurality of first leads are connected in one-to-one correspondence with the plurality of second leads, the plurality of first leads are located in the first display area, and the plurality of second leads are located in the peripheral area and are connected in one-to-one correspondence with the plurality of signal lines.
16. The display substrate according to claim 15, wherein the plurality of data lines have a first average line spacing in the second direction, the plurality of signal lines have a second average line spacing in the second direction, and the plurality of first leads have a third average line spacing in the second direction; wherein the third average line spacing is between the first average line spacing and the second average line spacing.
17. The display substrate according to claim 15, wherein the third average line spacing of the plurality of first leads is decreased, in the first direction, from a side close to the plurality of data lines to another side away from the plurality of data lines.
18. A display device, comprising the display substrate according to claim 1.
19. The display device according to claim 18, wherein the orthographic projections of at least some first light-emitting elements of the plurality of first light-emitting elements on the substrate surface of the base substrate have no overlap with orthographic projections of the plurality of first pixel driving circuits and the plurality of data lines on the substrate surface of the base substrate.
20. The display device according to claim 18, wherein the display substrate further comprises a plurality of second pixel driving circuits and a plurality of second light-emitting elements, wherein the plurality of second pixel driving circuits and the plurality of second light-emitting elements are located in the second display area, and the plurality of second pixel driving circuits are configured to drive the plurality of second light-emitting elements to emit light, the plurality of second pixel driving circuits are electrically connected with the plurality of data lines, and the plurality of data lines are configured to provide data signals to the plurality of second pixel driving circuits, and orthographic projections of the plurality of second light-emitting elements on the substrate surface of the base substrate are at least partly overlapped with orthographic projections of the plurality of second pixel driving circuits on the substrate surface of the base substrate.